

Hassan Ijaz

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Personal Statement

Data Analyst with a strong academic background in Data Analytics and Computer Science, and proven experience delivering end-to-end machine learning solutions across structured datasets and real-world domains. Skilled in designing predictive models with a focus on accuracy, interpretability, and reproducibility. Proficient in advanced Python, SQL, and PostgreSQL, feature engineering, and building analytics-ready datasets for BI tools including Excel, Power BI and Tableau. Experienced in integrating diverse data sources, including macroeconomic and financial market data, to enable systematic analysis while ensuring strong data quality and statistical rigour.

Education & Qualifications

MSc DATA ANALYTICS | UNIVERSITY OF HUDDERSFIELD, UK | MAY 2025 – MAY 2026

- Modules are Machine Learning | Statistical Analysis | Big Data Analysis | Predictive Modelling | Python for Data Science | Data Visualisation

BSc COMPUTER SCIENCE | UNIVERSITY OF MANAGEMENT AND TECHNOLOGY, PAKISTAN | SEP 2019 – SEP 2023

- Final year project: *Bid Bazar*, a full-stack auction site, later acquired by my supervisor.

Certifications

AI-POWERED DATA ANALYTICS | CODEWITHHARRY, ONLINE | NOV 2025 – APR 2026

- Advanced Python | SQL for analytics | advanced Excel | Git/GitHub | AI tools for analysts | probability and statistics | Power BI

Key Project

MSc DISSERTATION | PREDICTING CO2 EMISSIONS FROM UK HOUSING DATA | NOV 2025 – MAY 2026 | [GITHUB LINK](#)

Technologies: Python | Pandas | NumPy | Scikit-Learn | XGBoost | Random Forest, Gradient Boosting, SHAP, Matplotlib

Methods: Machine Learning | Predictive Modelling | Feature Engineering | Hyperparameter Tuning | Cross-Validation | Model Interpretability

- Architected an end-to-end predictive pipeline on 139,150 UK EPC residential records, applying IQR-based outlier removal to produce a refined dataset of 110,031 properties and engineering a structured feature framework from raw EPC description fields including roof insulation depth, wall and floor construction type, glazing properties, fuel group, and construction-year midpoint.
- Benchmarked four regression algorithms on an 80/20 train-test split, with XGBoost achieving R-squared = 0.96 and MAE = 0.20, outperforming Gradient Boosting (R-squared = 0.95) and Random Forest (R-squared = 0.93), confirmed through 5-fold cross-validation and Grid Search hyperparameter tuning.
- Built a parallel High/Low emitter classification model on a median-split target, achieving 96.2% accuracy and F1 = 0.96, complementing the continuous regression output with a binary risk-banding view useful for retrofit prioritisation policy.
- Applied SHAP-based interpretability analysis on the tuned XGBoost model to identify dominant emission drivers and covering exploratory analysis, model performance and feature attribution insights.

Bid Bazar | Full-Stack Online Auction Platform | Sep 2022 – Sep 2023

Technologies: JavaScript | Node.js | Express.js | MongoDB | MVC Architecture | RESTful APIs

Methods: Backend Development | Database Design | User Authentication | Real-Time Bidding Logic | Timer-Based Event Handling | Admin Dashboard Development

- Spearheaded backend implementation of BidBazar, a full-featured online auction portal designing and building user login, bid placement, timer-enabled auction closure, and complete administrative functionality on a JavaScript and MongoDB stack with an MVC architecture.
- Engineered a MongoDB document data layer with collection schemas covering users, listings, bids, and transaction logs, ensuring data integrity and supporting concurrent auction flows with efficient query ability across user and admin views.
- Built a session-based authentication and authorisation system separating buyer, seller, and administrator roles, alongside admin-side controls for listing approval, user management, and live auction monitoring.

- Delivered a timer-driven auction closure engine with automatic winner determination and concurrent bid handling logic the platform was subsequently acquired by the academic supervisor for continued development beyond the project's academic conclusion.

Skills & Achievements

- **Programming & Data Engineering:** Python | SQL | Pandas | NumPy | PostgreSQL | MySQL | ETL Pipelines | Data Cleaning | Feature Engineering | IQR Outlier Removal | Database Design | Git | GitHub
- **Machine Learning & Predictive Modelling:** Scikit-Learn | XGBoost | Random Forest | Gradient Boosting | GridSearchCV | Cross-Validation | SHAP | Hyperparameter Tuning | Time Series Analysis | Regression Analysis | Predictive Modelling | Hypothesis Testing | Root-Cause Analysis
- **Analytics, BI & Visualisation:** Advanced Excel | Power BI | Tableau | DAX | Matplotlib | Plotly | Interactive Dashboards | KPI Tracking & Reporting | Data Visualisation | Data Storytelling
- **Cloud Platforms:** Microsoft Azure | Google BigQuery
- **Financial & Quantitative Analysis:** Technical & Fundamental Analysis | Equity Earnings Analysis | Macroeconomic Data (CPI, GDP, Interest Rates, Inflation Rate, Central Bank Policy) | COT Positioning | Market Microstructure | Time Series Pattern Recognition | Risk Management | Position Sizing
- **Web Development:** HTML | CSS | JavaScript | React.js | Laravel (PHP) | RESTful APIs | Inertia.js
- **Stakeholder Engagement & Delivery:** Stakeholder Management | Stakeholder Reporting | Requirements Gathering | Presentation of Insights | Cross-Functional Collaboration | Agile/Scrum | Project Planning | Technical Documentation | Milestone Tracking | Data-Driven Documentation

Experience History

WEB DEVELOPER | AKVON IT SOLUTIONS, PAKISTAN | JUL 2023 – APR 2025

- Maintained client-facing web apps in React, Laravel/PHP, MySQL, and Inertia.js as part of a small dev team, contributing to feature work, bug fixes, and production support across multiple live projects.
- Built a session-based authentication system with role-based access for production accounts, separating permissions across admin, editor, and end-user tiers.
- Optimised the schema across relational databases and rewrote slow SQL on high-traffic endpoints, reducing query response times and improving stability under load.
- Developed REST APIs and the supporting data pipelines linking backend services to client-facing applications, cutting data transfer time by 30%.
- Owned database design decisions and stakeholder communication for an Australian client, translating business requirements directly into the technical implementation.

LINKS AVAILABLE ON REQUEST, SUBJECT TO COMPANY RESTRICTIONS.

FINANCIAL MARKET ANALYST | INDEPENDENT | 2021 – PRESENT

- Analysed financial markets independently since 2021 across crypto, forex, futures, options, and equities, covering both technical and fundamental sides of multi-asset analysis.
- Read price action, chart patterns, volume, and open interest across multiple timeframes to assess directional bias and market structure.
- Studied multiple years of historical setups to build intuition around how technical patterns typically resolve across different market regimes.
- Reviewed company earnings reports revenue, margins, guidance, and balance sheets to build informed views on individual stocks and broader sector trends.
- Tracked CPI, GDP, unemployment, central bank policy, interest rate decisions, and global news alongside COT reports to gauge institutional positioning and macro context.
- Produced monthly probability-weighted outlooks combining macro data, positioning, and technical setups across major asset classes.

Social Work

COMMUNITY VOLUNTEER | UNIVERSITY-AFFILIATED NGO, PAKISTAN | 2022 – 2023

- Led food drives and Ramadan iftar distribution for underprivileged families, coordinating donations and a team of volunteers from diverse donor and community backgrounds.